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👤 Immediate Joiner / Hybrid

Summary

Data Science undergraduate at **IIT Madras** (CGPA: **8.23/10**) building end-to-end AI/ML systems and production full-stack applications. Achieved **Kaggle Macro F1 of 0.954** via deep learning and transfer learning; experienced in LLM API integration, prompt engineering, and Python backend development (Flask, FastAPI, Celery). Seeking an **AI Full Stack / Backend Engineering Internship** to ship intelligent, API-driven products.

Skills

Languages: Python · JavaScript · SQL · Java
AI / ML & LLMs: PyTorch · Scikit-Learn · HuggingFace Transformers · OpenAI API · Gemini API · Prompt Engineering · TF-IDF · NLP · Transfer Learning · Feature Engineering · Ensemble Methods
Data & Backend: Pandas · NumPy · Flask · FastAPI · Django · REST APIs · JWT · Celery · Redis · SQLAlchemy
Frontend: Vue.js 3 · Pinia · Vue Router · HTML5 · CSS3 · Bootstrap 5
Databases & Tools: SQLite · PostgreSQL · MySQL · Git · GitHub · Linux · Jupyter Notebook · Weights & Biases

Projects

Messy Mashup Genre Classification 🔗 — Audio ML · Transfer Learning **Kaggle Leaderboard: Macro F1 = 0.954**
PyTorch · ResNet-50 · Audio Spectrogram Transformer · HuggingFace · Librosa · W&B

- Synthesized 10,000 noisy training mashups via ESC-50 noise injection and StratifiedGroupKFold (grouped by song ID) to close the distribution gap between clean stems and noisy test data — lifting baseline F1 from ~0.45 to **0.954**
- Fine-tuned **ResNet-50** (ImageNet) and **Audio Spectrogram Transformer** (AudioSet) on Log-Mel Spectrograms; applied Test-Time Augmentation (4 augmentations/chunk) and a weighted ensemble (60% AST + 40% ResNet) for the final leaderboard score

LLM Deployment AI Agent 🔗 — AI Engineering · Prompt Engineering · Backend Automation
FastAPI · OpenAI API · Gemini API · GitHub REST API · asyncio · Pydantic

- Engineered a prompt-driven REST API service that converts natural language instructions into structured, deployable HTML applications — using Pydantic schema validation to enforce JSON output instead of raw LLM text responses
- Orchestrated full deployment automation (GitHub repo creation, file commits, live deployment) via chained third-party API calls with asyncio for non-blocking, concurrent task handling

MedSyncPro 🔗 — Hospital Management System · Full Stack · Backend
Flask · Vue.js 3 (Composition API) · Pinia · Celery · Redis · SQLAlchemy · JWT · Bootstrap 5

- Architected a production-grade RBAC web application (Admin / Doctor / Patient) with a Flask REST backend, Vue.js 3 SPA, JWT authentication, and conflict-aware appointment booking with real-time slot validation
- Engineered a Celery + Redis async job pipeline delivering daily appointment reminders, automated monthly doctor performance reports (HTML email), and on-demand patient CSV exports — fully decoupled from the main API thread

Comment Category Prediction 🔗 — NLP · Ensemble Learning **Competition Macro F1 = 0.8295**
Scikit-Learn · LightGBM · TF-IDF · ColumnTransformer · SGD · LinearSVC

- Engineered 9 custom text and engagement features (caps_ratio, down_up_ratio, excl_count, etc.) unified in a ColumnTransformer pipeline with word n-grams (1,3) and char n-grams (3,5) on noisy, class-imbalanced real-world data
- Achieved **0.8295 Macro F1** via Voting Classifier (Bagging LR + Calibrated SVC + Tuned SGD) — outperforming standalone LightGBM (0.7386) and MLP (0.7460)

SmartParkr 🔗 — Vehicle Parking Management · Full Stack · Backend
Python · Flask · SQLite · SQLAlchemy · JavaScript · HTML / CSS · Bootstrap

- Designed a normalised 3-entity relational schema (Users, Slots, Bookings) with foreign-key constraints; built role-based REST APIs covering slot availability queries, booking conflict detection, and admin oversight dashboards
- Implemented session-based authentication with role-gated route protection and server-side input validation, ensuring data integrity at both the application and database layers

Education

B.S. in Data Science
Indian Institute of Technology Madras · CGPA: **8.23 / 10**
Higher Secondary Certificate (Class XII)
S.A. Jain Senior Model School · Grade: **9.0 / 10**

Sep 2024 – Present
Chennai, India
2023 – 2024
Ambala, Haryana